The documentation and process conversion measures necessary to comply with this document shall be completed by 18 February 2006.

INCH POUND

MIL-PRF-19500/317L
18 November 2005
SUPERSEDING
MIL-PRF-19500/317K
28 October 2004

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING, TYPES 2N2369A, 2N3227, 2N4449, 2N2369AU, 2N3227U, 2N2369AUA, 2N3227UA, 2N2369AUB, AND 2N3227UB, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, high speed switching transistors (including dual devices). Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500, and two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (TO-18) for 2N2369A and 2N3227, figure 2 (TO-46) for 2N4449, figure 3 for UB version, figure 4 for UA version, figure 5 for U version (dual devices), and figures 6 and 7 (JANC die).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Types	P_T	V_{CBO}	V_{EBO}	V_{CEO}	V_{CES}	$R_{\theta JA}$	T_J & T_{STG}
	W	V dc	V dc	V dc	V dc	°C/W	°C
2N2369A	0.36 (1)	40	4.5	15	40	325	-65
2N3227	0.36 (1)	40	6.0	20	40	325	to
2N4449	0.36 (1)	40	4.5	15	40	325	+200
All UA	0.5 (2)					210	
All UB	0.4 (3)					325	
All U	0.5 (4)					291	

(1) Derate linearly 2.06 mW/°C above T_A = +25°C.
(2) Derate linearly 4.76 mW/°C above T_C = +95°C.
(3) Derate linearly 3.08 mW/°C above T_C = +70°C.
(4) Derate linearly 3.44 mW/°C above T_A = +54.5°C.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

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* 1.4 Primary electrical characteristics at T_A = +25°C.

Type (1)	h_{FE2} (2) V_{CE} = 0.4 V dc I_C = 30 mA dc		h_{FE4} (2) V_{CE} = 1.0 V dc I_C = 100 mA dc		$\lvert h_{FE} \rvert$ V_{CE} = 10 V dc I_C = 10 mA dc f = 100 MHz		$V_{CE(sat)1}$ I_C = 10 mA dc I_B = 1 mA dc	t_{on} I_C = 10 mA dc I_{B1} = 3 mA dc I_{B2} = -1.5 mA dc	t_{off} I_C = 10 mA dc I_{B1} = 3 mA dc I_{B2} = -1.5 mA dc	t_s I_C = 10 mA dc I_{B1} = I_{B2} = 10 mA dc
	Min	Max	Min	Max	Min	Max	V dc Max	ns	ns	ns
2N2369A	30	120	20	120	5.0	10	0.20	12	18	13
2N3227	40	250	30	150	5.0	10	0.20	12	25	18
2N4449	30	120	20	120	5.0	10	0.20	12	18	13

(1) Electrical characteristics for the U, UA, and UB suffix devices are identical to the corresponding non-suffix device.
(2) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

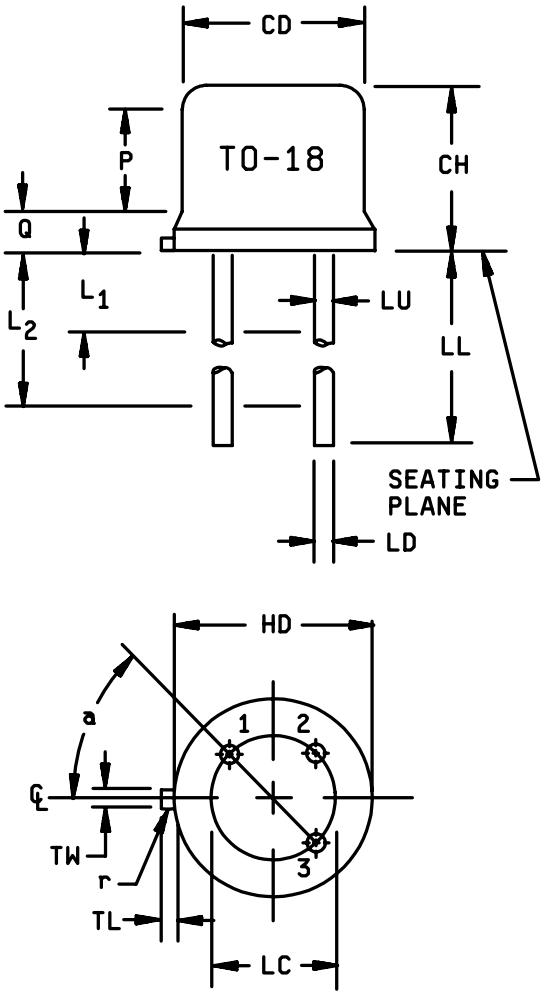
3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

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Ltr.	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8,13
LU	.016	.019	0.41	0.48	7,8
L1		.050		1.27	7,8
L2	.250		6.35		7,8
P	.100		2.54		5
Q		.030		0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r		.010		0.25	10
α	45° TP		45° TP		6

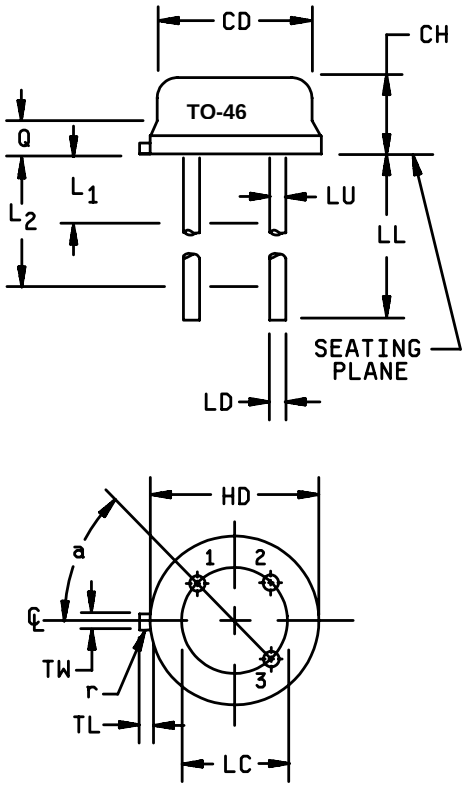


NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions TO-18 2N2369A and 2N3227.

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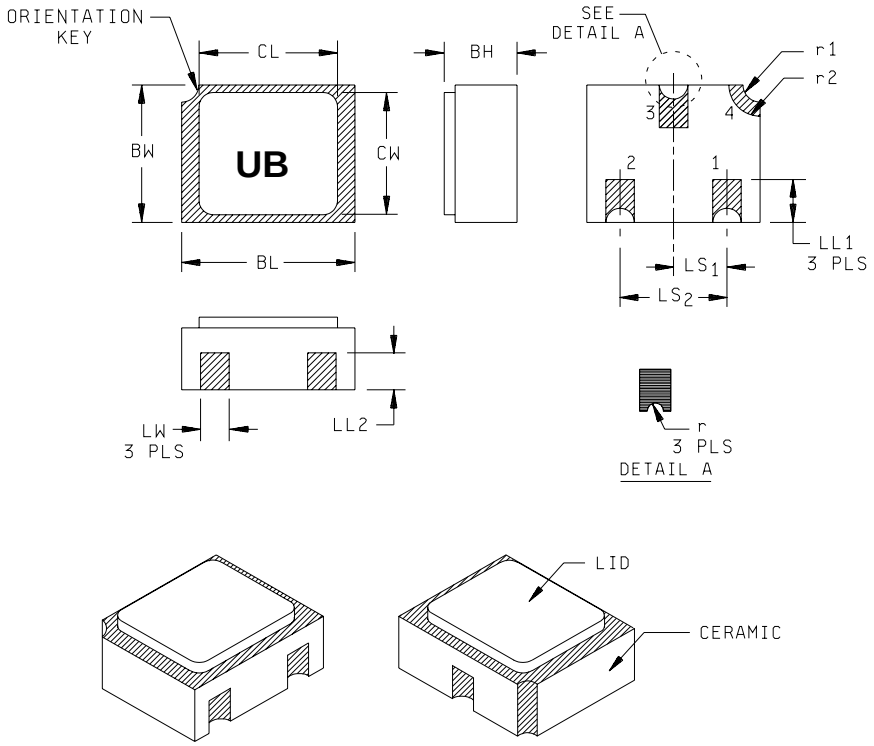
Ltr.	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.065	.085	1.65	2.16	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		5
LD	.016	.021	0.41	0.53	
LL	.500	1.750	12.70	44.45	6
LU	.016	.019	0.41	0.48	6
L1		.050		1.27	6
L2	.250		6.35		6
Q		.040		1.02	3
TL	.028	.048	0.71	1.22	8
TW	.036	.046	0.91	1.17	4
r		.010		0.25	9
α	45° TP		45° TP		5



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Leads at gauge plane .054 inch (1.37 mm) +.001 inch (0.03 mm) -.000 inch (0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
6. Symbol LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum.
7. Lead number three is electrically connected to case.
8. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
9. Symbol r applied to both inside corners of tab.
10. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
11. Lead 1 is emitter, lead 2 is base, and lead 3 is collector.

FIGURE 2. Physical dimensions – TO-46 2N4449.

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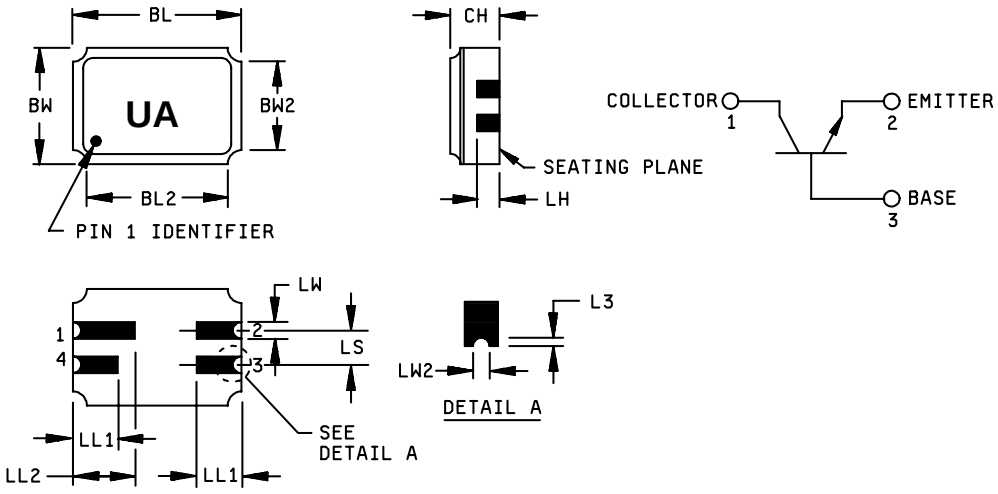


Ltr.	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
BH	.046	.056	1.17	1.42	
BL	.115	.128	2.92	3.25	
BW	.085	.108	2.16	2.74	
CL		.128		3.25	
CW		.108		2.74	
LL1	.022	.038	0.56	0.96	
LL2	.017	.035	0.43	0.89	
LS1	.036	.040	0.91	1.02	
LS2	.071	.079	1.81	2.01	
LW	.016	.024	0.41	0.61	
r		.008		.203	
r1		.012		.305	
r2		.022		.559	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas.
4. Lid material: Kovar.
5. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
6. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions - surface mount (UB version, 2N2369AUB, 2N3227UB).

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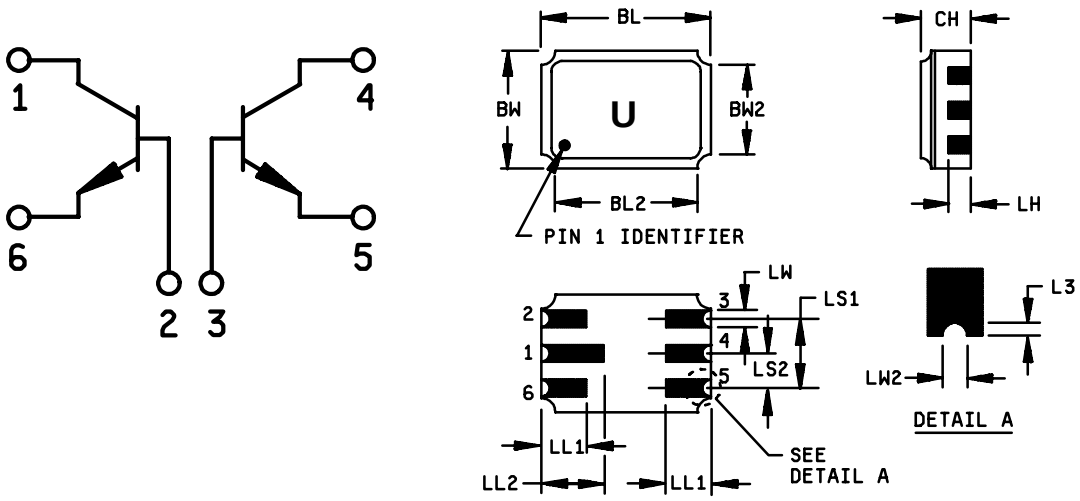
Ltr.	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.215	.225	5.46	5.71
BL_2		.225		5.71
BW	.145	.155	3.68	3.93
BW_2		.155		3.93
CH	.061	.075	1.55	1.90
L_3	.003	.007	0.08	0.18
LH	.029	.042	0.74	1.07
LL_1	.032	.048	0.81	1.22
LL_2	.072	.088	1.83	2.23
LS	.045	.055	1.14	1.39
LW	.022	.028	0.56	0.71
LW_2	.006	.022	0.15	0.56

Pin no.	1	2	3	4
Transistor	Collector	Emitter	Base	N/C

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimension CH controls the overall package thickness. When a window lid is used, dimension CH must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions LW2 minimum and L3 minimum and the appropriate castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on the bottom two layers, optional on the top ceramic layer.) Dimension LW2 maximum and L3 maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.
6. The co-planarity deviation of all terminal contact points, as defined by the device seating plane, shall not exceed .006 inch (0.15mm) for solder dipped leadless chip carriers.
7. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 4. Physical dimensions - surface mount (UA version, 2N2369AUA, 2N3227UA).

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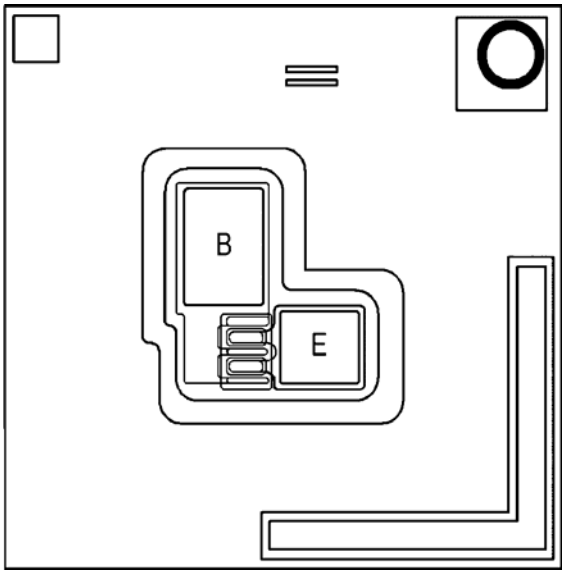
Ltr.	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.240	.250	6.10	6.35
BL_2		.250		6.35
BW	.165	.175	4.19	4.44
BW_2		.175		4.44
CH	.066	.080	1.68	2.03
L_3	.003	.007	0.08	0.18
LH	.026	.034	0.66	0.86
LL_1	.060	.070	1.52	1.78
LL_2	.082	.098	2.08	2.49
LS_1	.095	.105	2.41	2.67
LS_2	.045	.055	1.14	1.39
LW	.022	.028	0.56	0.71
LW_2	.006	.022	0.15	0.56

Pin No.	1	2	3	4	5	6
Transistor	Collector no. 1	Base no. 1	Base no. 2	Collector no. 2	Emitter no. 2	Emitter no. 1

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimension CH controls the overall package thickness. When a window lid is used, dimension CH must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions LW2 minimum and L3 minimum and the appropriate castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on the bottom two layers, optional on the top ceramic layer.) Dimension LW2 maximum and L3 maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.
6. The co-planarity deviation of all terminal contact points, as defined by the device seating plane, shall not exceed .006 inch (0.15mm) for solder dipped leadless chip carriers.
7. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 5. Physical dimensions - surface mount (dual transistors, U version only, 2N2369AU, 2N3227U).

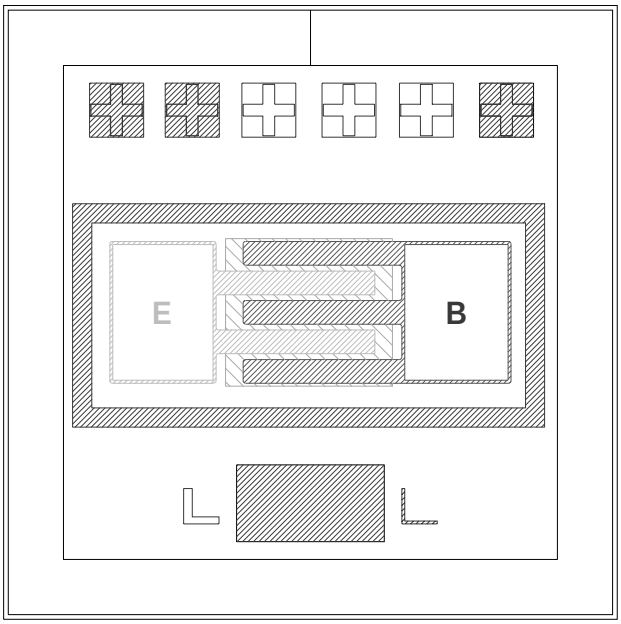
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Die size:	20 x 20 mils ± 2 mils.
Die thickness:	10 ± 1.5 mils nominal.
Top metal:	Aluminum 10,000Å minimum, 12,000Å nominal.
Back metal:	a. Al/Ti/Ni/Ag 12kÅ/3kÅ/7kÅ/7kÅ min.,15kÅ/5kÅ/10kÅ/10kÅ nominal. b. Gold 2,500Å minimum, 3,000Å nominal. c. Eutectic Mount – No Gold.
Backside:	Collector.
Bonding pad:	B = 4 x 4.5 mils, E = 4.5 x 5 mils.

FIGURE 6. JANHC and JANKC A-version die dimensions - 2N2369A.

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Die size:	16 mils x 16 mils.
Die thickness:	8 mils ±1.6 mils.
Base pad:	3.6 mils x 2.8 mils.
Emitter pad:	3.6 mils x 2.8 mils.
Back metal:	Gold, 6500 ±1950 Ang.
Top metal:	Aluminum, 17500 ±2500 Ang.
Back side:	Collector.
Glassivation:	SiO_2, 7500 ±1500 Ang.

FIGURE 7. JANHC and JANKC B-version die dimensions - 2N2369A, 2N3227.

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3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I herein. When a particular device is specified, the limit applies to all package types (e.g., 2N2369A, 2N2369AU, 2N2369AUA, and 2N2369AUB).

3.6 Electrical test requirements. The electrical test requirements shall be specified in table I. When a particular device is specified, the limit applies to all package types (e.g., 2N2369A, 2N2369AU, 2N2369AUA, and 2N2369AUB).

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "AUB" suffix may also be omitted.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table VI of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
7	Optional.	Optional.
9	I_{CES} and h_{FE3}	Not applicable
11	I_{CES}; h_{FE3}; ΔI_{CES} = 100 percent of initial value or 25 nA dc, whichever is greater. Δh_{FE3} = ±15 percent of initial value.	I_{CES}, h_{FE3}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CES} = 100 percent of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = ±15 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CES} = 100 percent of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = ±15 percent of initial value.
14	Required.	Required.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CE} = 5 - 15 V dc, P_D = 360 mW, T_A = room ambient as defined in 4.5 of MIL-STD-750. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions, etc.) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

* 4.3.2. Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with appendix G of MIL-PRF-19500. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.3.3 Thermal impedance (ΔV_{BE} measurements). The ΔV_{BE} measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining V_H, V_{CE}, I_M, I_H, t_H, t_{SW}, and t_{MD}. The ΔV_{BE} limit used in screen 3c of 4.3 herein and table I, subgroup 2 shall be set statistically by the supplier over several die lots and submitted to the qualifying activity for approval. See group E, subgroup 4 herein.

* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed in accordance with MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta requirements shall be in accordance with table II herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 12 V dc, t_{ON} = t_{OFF} = 3 minutes, $P_{D(ON)}$ = P_D max rated, see 1.3; $P_{D(OFF)}$ = 0.
B5	1027	V_{CB} = 12 V dc. $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). Option 1: 96 hrs min, sample size in accordance with table VIa of MIL-PRF-19500, adjust T_A to achieve T_J = +275°C minimum. Option 2: 216 hrs min., sample size = 45, c = 0; adjust T_A to achieve T_J = +225°C minimum. (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)

* 4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV) Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double the sample size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new assembly lot option is exercised, the failed assembly lot shall be scrapped.

	Step	Method	Condition
*	1	1026	Steady-state life: 1,000 hours, V_{CB} = 5 - 15 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 100 percent of maximum rated P_T as defined in 1.3, in addition, adjust T_A to achieve T_J. n = 45 devices, c = 0. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
	2	1048	Blocking life: T_A = +150°, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
	3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta requirements shall be in accordance with table II herein.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

	Subgroup	Method	Condition
	C2	2036	Test condition E; not applicable for U, UA, and UB devices.
*	C6	1026	1,000 hours at V_{CB} = 5 - 15 V dc. Maximum rated power (see 1.3) shall be applied to the device and ambient temperature shall be adjusted to achieve $T_J \geq +150°C$. The sample size may be increased and the test time decreased so long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. n = 45, c = 0.

4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for U, UA, and UB devices.
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2. Delta requirements shall be in accordance with table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

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* TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 3/						
Visual and mechanical 4/ examination	2071	n = 45 devices, c = 0				
Solderability 4/ 5/	2026	n = 15 leads, c = 0				
Resistance to solvents 4/ 5/ 6/	1022	n = 15 devices, c = 0				
Temp cycling 4/ 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Hermetic seal 5/ Fine leak Gross leak	1071	n = 22 devices, c = 0 Test conditions G or H Test conditions C or D				
Electrical measurements 5/		Table I, subgroup 2				
Bond strength 4/ 5/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = 300°C at t = 2 hrs; n = 11 wires, c = 0				
Decap internal visual 5/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance 7/	3131	See 4.3.3	ΔV_{BE}			mV
Collector to base cutoff current	3036	V_{CB} = 40 V dc	I_{CBO1}		10	µA dc
Emitter to base cutoff current 2N2369A, 2N4449 2N3227	3061	 V_{EB} = 4.5 V dc V_{EB} = 6.0 V dc	I_{EBO1}		10	µA dc
Breakdown voltage, collector to emitter 2N2369A, 2N4449 2N3227	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	 15 20		 V dc V dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} = 20 V dc	I_{CES}		0.4	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 32 V dc	I_{CBO2}		0.2	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		0.25	µA dc
Forward-current transfer ratio 2N2369A, 2N4449, 2N3227	3076	V_{CE} = 0.35 V dc; I_C = 10 mA dc pulsed (see 4.5.1)	h_{FE1}	 40 70	 120 250	
Forward-current transfer ratio 2N2369A, 2N4449 2N3227	3076	V_{CE} = 0.4 V dc; I_C = 30 mA dc pulsed (see 4.5.1)	h_{FE2}	 30 40	 120 250	

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc pulsed (see 4.5.1)	h_{FE3}			
2N2369A, 2N4449				40	120	
2N3227				75	300	
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 100 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N2369A, 2N4449				20	120	
2N3227				30	150	
Collector-emitter saturation voltage	3071	I_C = 10 mA dc; I_B = 1.0 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.20	V dc
Collector-emitter saturation voltage	3071	I_C = 30 mA dc; I_B = 3.0 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.25	V dc
Collector-emitter saturation voltage	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		0.45	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.70	0.85	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 30 mA dc; I_B = 3.0 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		0.90	
Base-emitter saturation voltage	3066	Test condition A; I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)3}$	0.80	1.20	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 20 V dc	I_{CBO2}		30	µA dc
High temperature operation		T_A = +125°C				
Collector to base cutoff current	3041	Bias condition A; V_{CE} = 10 V dc; V_{BE} = 0.25 V dc	I_{CEX2}		30	µA dc
Collector - emitter voltage saturated	3071	I_C = 10 mA dc I_B = 1.0 mA dc	$V_{CE(sat)4}$		0.3	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)4}$			
2N2369A, 2N4449				0.59		V dc
2N3227				0.50		V dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc pulsed (see 4.5.1)	h_{FE5}			
2N2369A, 2N4449				20		
2N3227				40		
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)5}$		1.02	V dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 10 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	5.0	10	
Open circuit output capacitance	3236	V_{CB} = 5 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		4.0	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}			
2N2369A, 2N4449					5.0	pF
2N3227					4.0	pF
Charge storage time		I_C = 10 mA dc; I_{B1} = 10 mA dc; I_{B2} = 10 mA dc; (see figure 8)	t_S			
2N2369A, 2N4449					13	ns
2N3227					18	ns
Turn-on time		I_C = 10 mA dc; I_{B1} = 3.0 mA dc; I_{B2} = -1.5 mA dc; (see figure 9)	t_{on}		12	ns
Turn-off time		I_C = 10 mA dc; I_{B1} = 3.0 mA dc; I_{B2} = -1.5 mA dc; (see figure 9)	t_{off}			
2N2369A, 2N4449					18	ns
2N3227					25	ns

1/ For sampling (unless otherwise specified) plan see MIL-PRF-19500.
2/ Electrical characteristics for the U, UA, and UB suffix devices are identical to the corresponding non-suffix device.
3/ For resubmission of failed subgroup 1, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
4/ Separate samples may be used.
5/ Not required for JANS devices.
6/ Not required for laser marked devices.
7/ This test required for the following end-point measurements only:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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* TABLE II. Groups B, C, and E delta measurements.

Step	Inspection 1/ 2/ 3/ 4/	MIL-STD-750		Symbol	Limit
		Method	Conditions		
1.	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	Δh_{FE3}	±25 percent change from initial value.
2.	Collector - emitter and resistance	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$\Delta V_{CE(sat)1}$	±50 mV change from previous measured value.
3.	Collector - emitter cutoff current	3041	Bias condition C; V_{CE} = 20 V dc;	ΔI_{CES}	100 percent of initial value or 25 nA dc, whichever is greater.

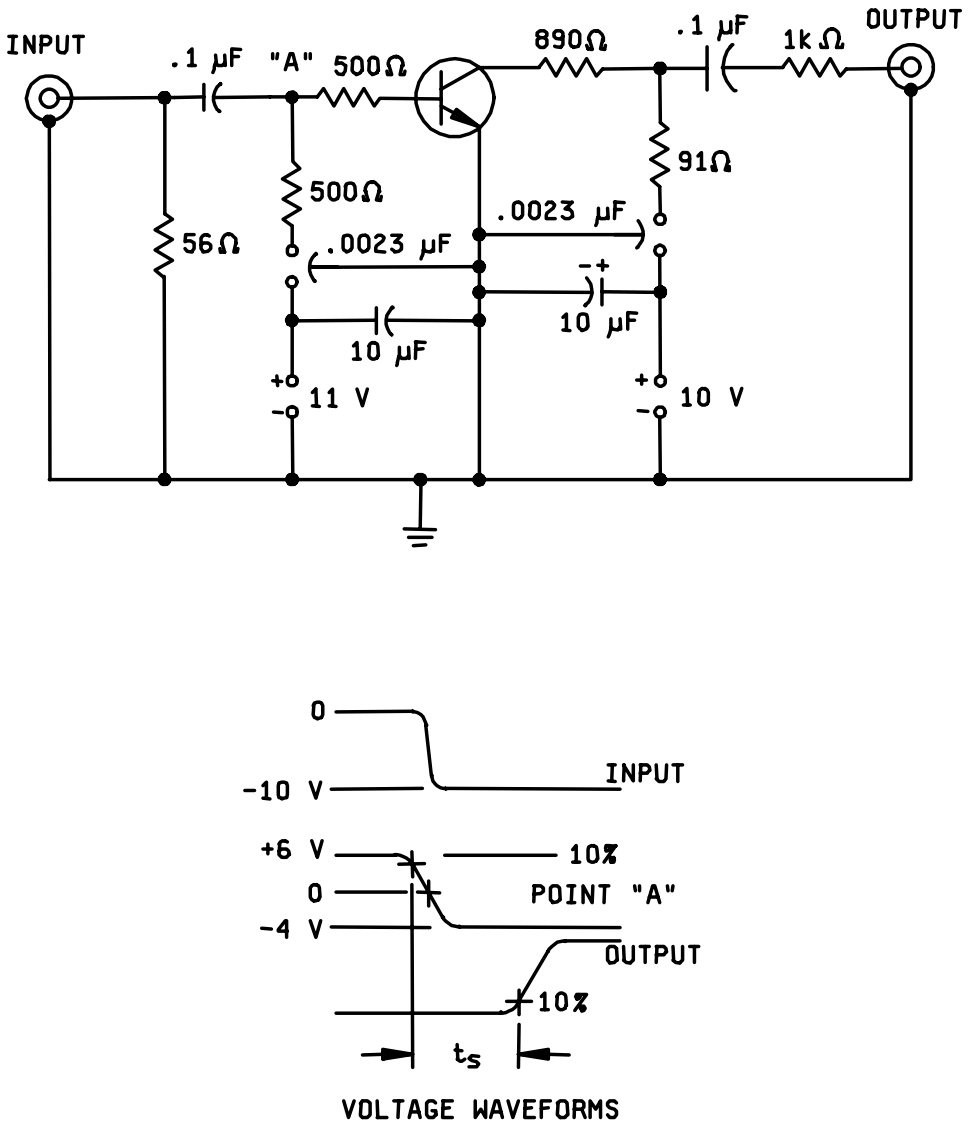
1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows: Subgroups B4 and B5, see table II herein, steps 1, 2, and 3.
2/ The delta measurements (JAN, JANTX and JANTXV) of 4.4.2.2 herein are as follows: Steps 1, 2, and 3, see table II herein, steps 1 and 3.
3/ The delta measurements for table VII of MIL-PRF-19500 and 4.4.3.1 are as follows: Subgroup 6, see table II herein, steps 2 and 3 (for JANS).
4/ The delta measurements for table IX of MIL-PRF-19500 and table II herein are as follows: Subgroups 1 and 2, see table II herein, all steps.

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* TABLE III. Group E inspection (all quality levels) – for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
Electrical measurements		See table I, subgroup 2 and table II herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc; 6,000 cycles, $t_{ON} = t_{OFF}$ = 3 minutes, $P_{DON} = P_D$ max rated in accordance with 1.3; P_{DOFF} = 0.	
Electrical measurements		See table I, subgroup 2 and table II herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		Each supplier shall submit their qual-lot average and design maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	
Subgroup 5			
Not applicable			
Subgroup 6			3 devices
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B	

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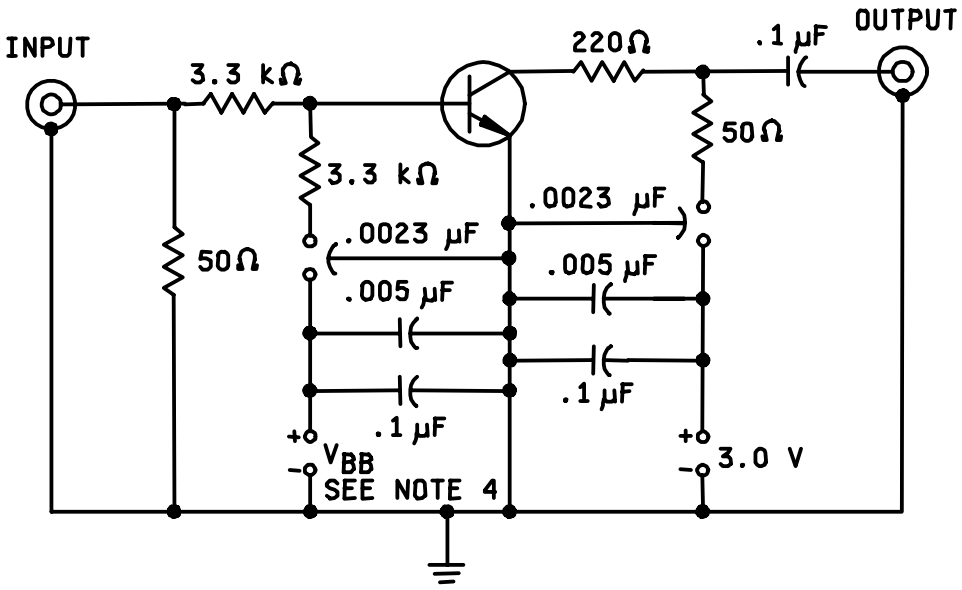
VOLTAGE WAVEFORMS

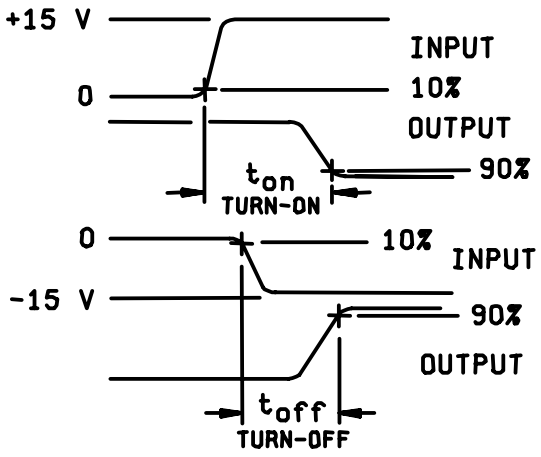
NOTES:

1. All capacitance in µF.
2. The input waveforms for each circuit are supplied by a pulse generator with the following characteristics: Z_{OUT} = 50Ω, $t_r \leq 1$ ns, PW ≥ 300 ns, duty cycle ≤ 2 percent.
3. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq .1$ ns, Z_{IN} = 50Ω.

FIGURE 8. Charge storage time test circuit.

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VOLTAGE WAVEFORMS

NOTES:

1. All capacitance in µF.
2. The input waveforms for each circuit are supplied by a pulse generator with the following characteristics: Z_{OUT} = 50Ω, $t_r \leq 1$ ns, PW ≥ 300 ns, duty cycle ≤ 2 percent.
3. Input and output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq .1$ ns, Z_{IN} = 50Ω.
4. V_{BB} = -3.0 V for t_{on}, +12.0 V for t_{off}.

FIGURE 9. Turn-on and turn-off time test circuit.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N2369A) will be identified on the QML.

Die ordering information		
PIN	Manufacturer	
	43611	34156
2N2369A 2N3227	JANHCA2N2369A, JANKCA2N2369A JANHCA2N3227, JANKCA2N3227	JANHCB2N2369A, JANKCB2N2369A JANHCB2N3227, JANKCB2N3227

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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